

James Lewis - S2 - 29

Experiment 07

server.py

```
import socket # Importing socket module for networking
import threading # Importing threading module to handle multiple clients

# Define server configuration
PORT = 5050 # Port number on which the server will listen
SERVER = socket.gethostbyname(socket.gethostname()) # Automatically get local IP address
ADDR = (SERVER, PORT) # Combine IP and port into a tuple
FORMAT = 'utf-8' # Encoding format for messages
HEADER = 64 # Fixed header size to store message length
DISCONNECT_MESSAGE = '!DISCONNECT' # Command to disconnect the client

# Create a socket for the server (IPv4, TCP)
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(ADDR) # Bind the server to the specified address and port

def clientHandling(connection, address):
    """
    Handles communication with a connected client.

    Parameters:
    - connection: The socket object for client-server communication
    - address: The client's address (IP and port)
    """
    print(f'[NEW CONNECTION] {address} connected')
    isConnected = True

    while isConnected:
        # Receive the message length (first HEADER bytes)
        messageLength = connection.recv(HEADER).decode(FORMAT)
```

```

        if messageLength:
            messageLength = int(messageLength) # Convert length to
integer
            message = connection.recv(messageLength).decode(FORMAT) #
Receive actual message

            if message == DISCONNECT_MESSAGE: # Check if the client wants
to disconnect
                isConnected = False

            print(f'[{address}] {message}') # Print the received message
            connection.send('Message Received'.encode(FORMAT)) #
Acknowledge receipt

        connection.close() # Close the client connection when loop ends

def start():
    """
    Starts the server and listens for incoming connections.
    """
    server.listen() # Start listening for client connections
    print(f'[LISTENING] Server is listening on {SERVER}')

    while True:
        # Accept a new client connection
        connection, address = server.accept()

        # Create a new thread for handling this client
        thread = threading.Thread(target=clientHandling, args=(connection,
address))
        thread.start()

        # Print the number of active connections (excluding the main
thread)
        print(f'[ACTIVE CONNECTIONS] {threading.active_count() - 1}')

# Start the server
print('[SERVER STARTING] Server has been started')
start()

```

client.py

```
import socket

PORT = 5050
SERVER = '192.168.137.184' # Your IP Address
ADDR = (SERVER, PORT)
FORMAT = 'utf-8'
HEADER = 64
DISCONNECT_MESSAGE = '!DISCONNECT'

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(ADDR)

def send(messages):
    message = messages.encode(FORMAT)
    messageLength = len(message)
    sendLength = str(messageLength).encode(FORMAT)
    sendLength += b' ' * (HEADER - len(sendLength))
    client.send(sendLength)
    client.send(message)
    print(client.recv(2045).decode(FORMAT))

if __name__ == '__main__':
    # Send "Hello World" message
    send("Hello World")

    # Disconnect
    send(DISCONNECT_MESSAGE)
```

Implementation:

```
D:\Experiment07>py server.py
[SERVER STARTING] Server has been started
[LISTENING] Server is listening on 192.168.137.184
[NEW CONNECTION] ('192.168.137.184', 57933) connected
[ACTIVE CONNECTIONS] 1
[('192.168.137.184', 57933)] Hello World
[('192.168.137.184', 57933)] !DISCONNECT
```

```
Microsoft Windows [Version 10.0.26120.3360]
(c) Microsoft Corporation. All rights reserved.

C:\Users\James Lewis>cd..

C:\Users>ce.
'ce.' is not recognized as an internal or external command,
operable program or batch file.

C:\Users>cd..

C:\>cd /D D:

D:\>cd "D:\Experiment07"

D:\Experiment07>py client.py
Traceback (most recent call last):
  File "D:\Experiment07\client.py", line 24, in <module>
    send("Hello World")
    ~~~~~^^^^^^^^^^^^^^^^^^^^^^^^
  File "D:\Experiment07\client.py", line 20, in send
    print(client.recv(2045).decode(FORMAT))
           ~~~~~^^^^^^^^^^^^^^^^^^^^^^^^
ConnectionResetError: [WinError 10054] An existing connection was forcibly closed by the remote host

D:\Experiment07>py client.py
Message Received
Message Received

D:\Experiment07>
```